Complex Coefficients, and Phases

(Conceptual Foundations before Time Evolution)

This section aims to clarify **why quantum states are described using complex coefficients**, what **phases** mean physically, and how this naturally leads to **unitary transformations** and the **Hamiltonian**, *before* introducing time evolution explicitly.

In quantum mechanics, a **quantum state** is a mathematical object that allows us to **predict probabilities of measurement outcomes**.

Consider a system with two possible outcomes for a given measurement:

$|u_1\rangle \quad , \quad |u_2\rangle$

A general state is written as:

$|\psi\rangle = c_1 |u_1\rangle + c_2 |u_2\rangle$

At this stage:

- c_1 and c_2 are numbers called **probability amplitudes**
- The state is a **superposition** of possible outcomes

According to **Born's rule**:

$P(u_1) = |c_1|^2 \quad , \quad P(u_2) = |c_2|^2$

This correctly gives measurement probabilities.

However, **probabilities alone do not fully characterize the quantum state**.
Different states may yield the same probabilities for one measurement, but produce different results when measuring another observable.

Therefore, the state must contain **more information than probabilities alone**.

The most general form of the coefficients is: $\quad c_1 = |c_1|e^{i\theta_1}, \quad c_2 = |c_2|e^{i\theta_2}$

The probabilities depend only on the magnitudes:

$$|c_1|^2, \quad |c_2|^2$$

But the **phases** θ_1, θ_2 carry additional information about the state.

We can factor out a common phase:

$|\psi\rangle = e^{i\theta_1} \left(|c_1| |u_1\rangle + |c_2| e^{i(\theta_2 - \theta_1)} |u_2\rangle \right)$

Any measurable probability is of the form:

$| \langle \phi | \psi \rangle |^2$

The global phase factor $e^{i\theta_1}$ always cancels out.

Conclusion:
Two states that differ only by a global phase represent the **same physical state**.

The remaining phase difference:

$\Delta\theta = \theta_2 - \theta_1$

is called the **relative phase**.

Example:

$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|u_1\rangle + |u_2\rangle)$

$|\psi_2\rangle = \frac{1}{\sqrt{2}}(|u_1\rangle - |u_2\rangle)$

These states:

- have identical probabilities in the $|u_1\rangle, |u_2\rangle$ basis
- are physically **distinct**

In another measurement basis, they yield different outcomes due to **interference terms**.

Relative phases affect measurable quantities.

But why are these coefficients complex numbers? Because:

- phases must vary **continuously**
- interference effects depend on smooth phase changes
- probabilities are computed using modulus squared

Complex numbers are the **minimal mathematical structure** that allows:

- amplitudes and phases
- continuous transformations
- consistent probabilistic interpretation

This is not a convention, but a **physical necessity**.

From the above, we conclude:

- A quantum state is a **normalized vector** in a complex vector space
- States differing by a global phase are physically identical

This leads naturally to the concept of **projective Hilbert space**.

Suppose the state undergoes some transformation:

$|\psi\rangle \to |\psi'\rangle$

This transformation must:

- preserve normalization
- preserve probabilities
- preserve relative phases

The only linear transformations with these properties are **unitary operators**:

$|\psi'\rangle = U|\psi\rangle \quad , \quad U^\dagger U = I$

Thus, **unitarity is not postulated arbitrarily** — it is imposed by the structure of quantum states and measurements.

Any continuous unitary transformation can be written as:

$U(t) = e^{-iHt/\hbar}$

where:

- H is a Hermitian operator
- H generates changes in relative phases

This operator is called the **Hamiltonian**.

Key idea:

The Hamiltonian emerges as the generator of physically meaningful phase evolution, not merely as an energy operator.

Conceptual Summary

- Quantum states require more than probabilities → phases are essential
- Global phase is unobservable
- Relative phases encode physical information
- Complex numbers are unavoidable
- Valid state transformations must be unitary
- The Hamiltonian generates unitary evolution by controlling phase relations

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This framework prepares the ground for understanding **time-independent Schrödinger equations, stationary states, and quantum dynamics**, which will be introduced in the next section.

For a two-level system, this structure is exactly what defines a qubit, whose physical information is entirely e